HOSPITAL APPOINTMENT MANAGEMENT SYSTEM

Name: ARNAB DE

Stream: CSE(IOT)

Roll No: 02; Sec: A

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Enrollment No: 12024052019014

Course: Advanced Programming (PCCCS495)

1. INTRODUCTION:

Managing hospital appointments manually is inefficient and time-consuming. Patients often face delays and confusion in scheduling. This project provides a console-based system to manage patients, doctors, and appointments efficiently.

2. OBJECTIVES:

- To manage patient records
- To manage doctor details
- To schedule appointments
- To reduce manual work
- To apply OOP concepts

3. SYSTEM FEATURES:

- Add Patient
- View Doctors
- Book Appointment
- View Appointments
- Cancel Appointment

4. OOP CONCEPTS USED:

I. Abstraction: User class defines common attributes.
II. Inheritance: Patient class extends User class.
III. Encapsulation: Private variables are used with proper access methods.
IV. Polymorphism: Used in method handling and flexibility.
V. Exception Handling: Used to handle invalid inputs.
VI. Collections: ArrayList used to store data.

5. SYSTEM ARCHITECTURE:

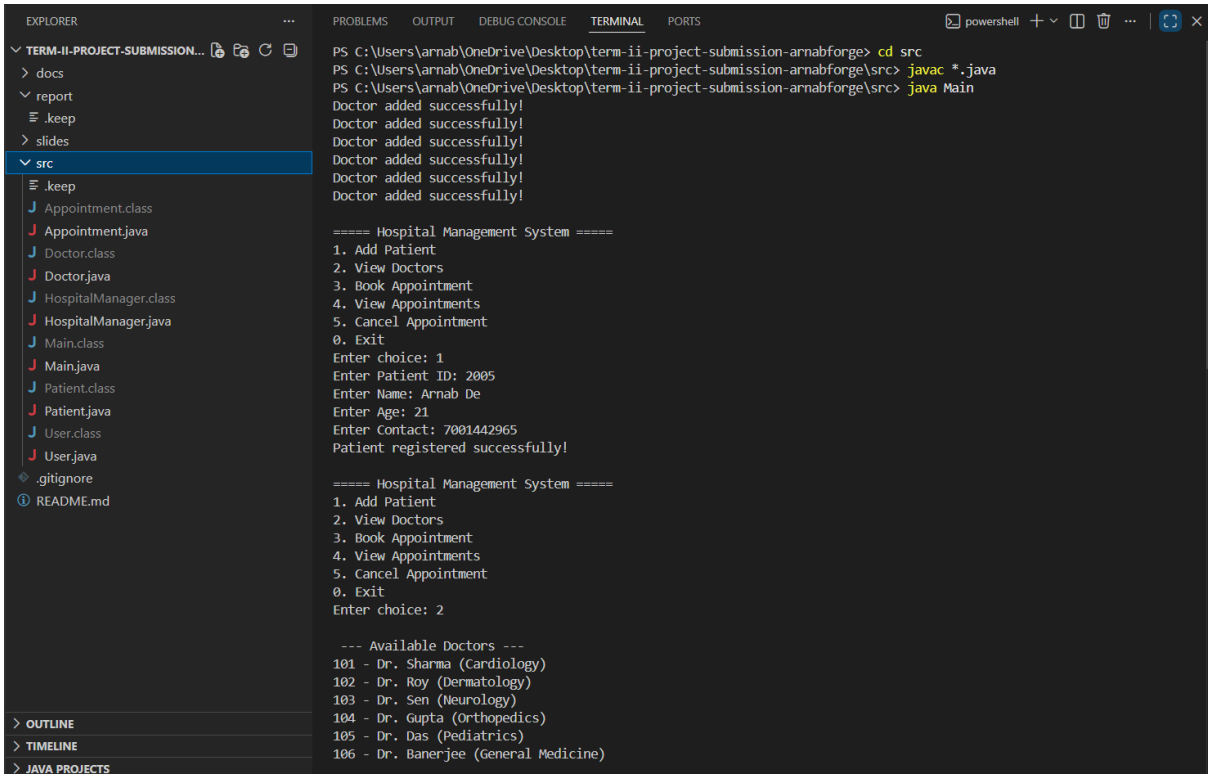
The system is divided into:

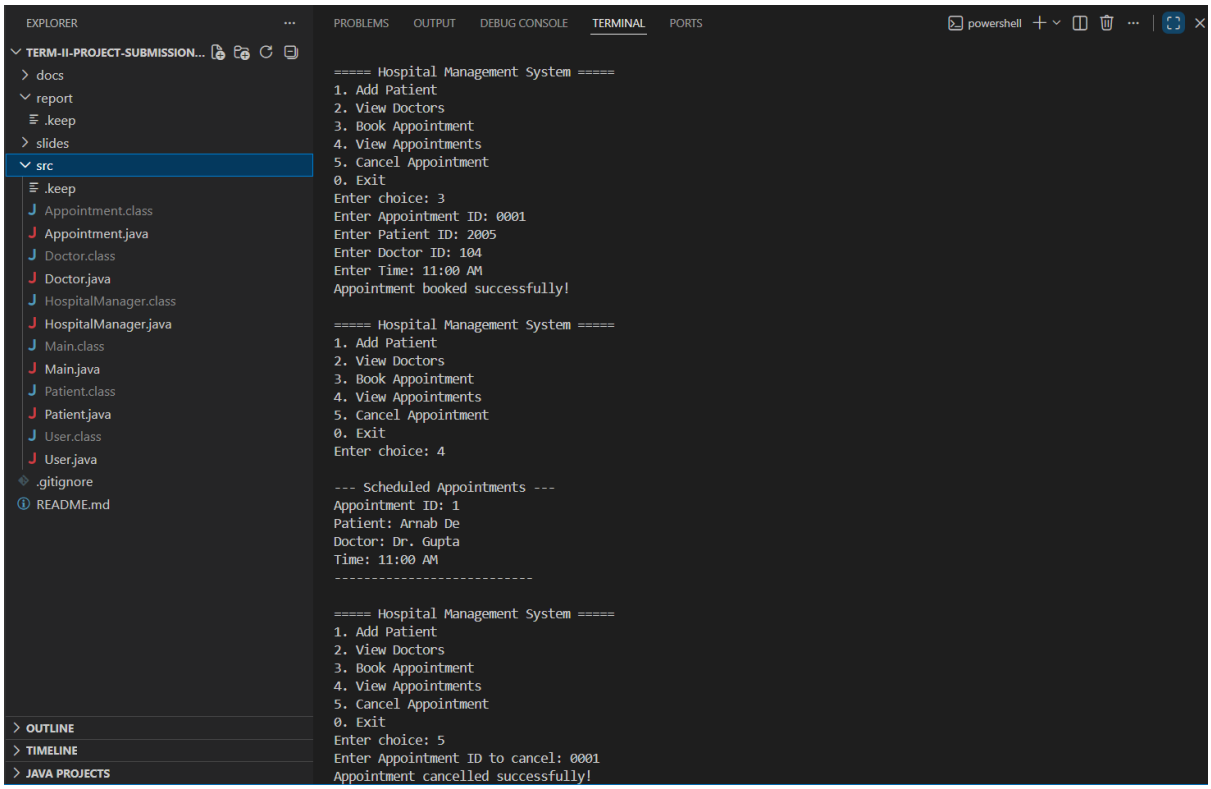
- User (base class)
- Patient
- Doctor
- Appointment
- Hospital Manager
- Main class

6. HOW TO RUN:

1) Open terminal
2) Go to src folder
3) Compile: javac *.java
4) Run: java Main

7. OUTPUT:


EXPLORER
TERM-II-PROJECT-SUBMISSION...
docs
report
.keep
slides
src
.keep
Appointment.class
Appointment.java
Doctor.class
Doctor.java
HospitalManager.class
HospitalManager.java
Main.class
Main.java
Patient.class
Patient.java
User.class
User.java
.gitignore
README.md
OUTLINE
TIMELINE
JAVA PROJECTS
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
powershell
PS C:\Users\arnab\OneDrive\Desktop\term-ii-project-submission-arnabforge> cd src
PS C:\Users\arnab\OneDrive\Desktop\term-ii-project-submission-arnabforge\src> javac *.java
PS C:\Users\arnab\OneDrive\Desktop\term-ii-project-submission-arnabforge\src> java Main
Doctor added successfully!
Doctor added successfully!
Doctor added successfully!
Doctor added successfully!
Doctor added successfully!
Doctor added successfully!
===== Hospital Management System =====
1. Add Patient
2. View Doctors
3. Book Appointment
4. View Appointments
5. Cancel Appointment
0. Exit
Enter choice: 1
Enter Patient ID: 2005
Enter Name: Arnab De
Enter Age: 21
Enter Contact: 7001442965
Patient registered successfully!
===== Hospital Management System =====
1. Add Patient
2. View Doctors
3. Book Appointment
4. View Appointments
5. Cancel Appointment
0. Exit
Enter choice: 2
--- Available Doctors ---
101 - Dr. Sharma (Cardiology)
102 - Dr. Roy (Dermatology)
103 - Dr. Sen (Neurology)
104 - Dr. Gupta (Orthopedics)
105 - Dr. Das (Pediatrics)
106 - Dr. Banerjee (General Medicine)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===== Hospital Management System =====
1. Add Patient
2. View Doctors
3. Book Appointment
4. View Appointments
5. Cancel Appointment
0. Exit
Enter choice: 3
Enter Appointment ID: 0001
Enter Patient ID: 2005
Enter Doctor ID: 104
Enter Time: 11:00 AM
Appointment booked successfully!
===== Hospital Management System =====
1. Add Patient
2. View Doctors
3. Book Appointment
4. View Appointments
5. Cancel Appointment
0. Exit
Enter choice: 4
--- Scheduled Appointments ---
Appointment ID: 1
Patient: Arnab De
Doctor: Dr. Gupta
Time: 11:00 AM

===== Hospital Management System =====
1. Add Patient
2. View Doctors
3. Book Appointment
4. View Appointments
5. Cancel Appointment
0. Exit
Enter choice: 5
Enter Appointment ID to cancel: 0001
Appointment cancelled successfully!

8. LIMITATIONS:

1) Console-based system
2) No database storage

9. FUTURE SCOPE:

1) Add GUI (Swing/JavaFX)
2) Add database (MySQL)

10. CONCLUSION:

This project demonstrates the use of OOP concepts in a real-world application.

It improves efficiency in managing hospital appointments.